Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

CH407 - ORGANIC CHEMISTRY - II

Time : 3 hrs **Max Marks :75**

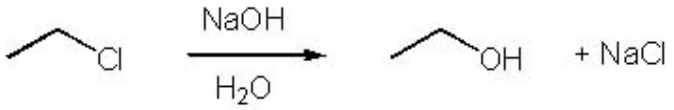
SECTION – A (20X1=20)

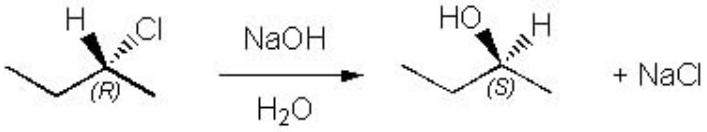
Answer **ALL** Questions

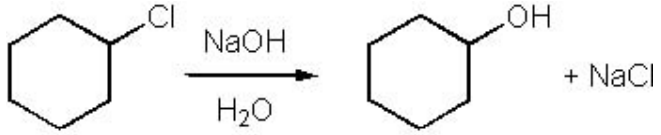
I. Choose the correct answer

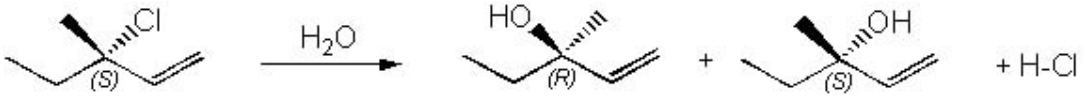
1. Which among the following is a good nucleophile?
   a) NH_3 b) HBr c) Br_2 d) BH_3

2. Which of the following is an SN1reaction?

a) 

b) 

c) 

d) 

3. An organic compound is aromatic if
   a) Pi-bonding molecular orbitals are fully filled
   b) Non-bonding pi molecular orbitals are fully filled
   c) Non-bonding pi molecular orbitals are half filled
   d) (a) and (b) are correct if molecule is flat

4. A new C-C bond is formed in
   a) Cannizzaro's reaction b) Friedel Crafts reaction
   c) Clemmensen reduction d) None of the above

5. An aromatic ether is not cleaved by HI even at 525 K. The compound is
   a) $C_6H_5OCH_3$ b) $C_6H_5OC_6H_5$ c) $C_6H_5OC_3H_7$ d) Tetrahydrofuran

6. Conversion of alcohols into chloro-compounds by reaction with SOCl2 follows the order
   a) $1^o > 2^o > 3^o$ b) $3^o > 2^o > 1^o$ c) $2^o > 1^o > 3^o$ d) $1^o = 2^o > 3^o$

7. Which among the following can reduce a ketene into hydrocarbon?
   a) $LiAlH_4$ b) $NaBH_4$ c) H_2/Ni d) N_2H_4/KOH

8. Aldol condensation of acetaldehyde involves the formation of which of the following intermediate?
   a) Acetate ion b) A carbanion c) A carbonium ion d) A free radical

9. Strongest acid among the following is
   a) CI_3COOH b) CBr_3COOH c) CCl_3COOH d) CF_3COOH

10. Which of the following would be expected to be most highly ionised in water?
    a) $CH_3CHClCH_2CH_2COOH$ b) $CH_3CHClCH_2COOH$
    c) $CH_3CH_2CCl_2COOH$ d) $CH_3CH_2CHClCOOH$

II. Fill up the blanks

11. The reaction of 3-chloro-2,2-dimethylbutane and sodium iodide yields _____

12. The structure of azulene is _____

13. _____ is called picric acid

14. Aldehydes and ketones having alpha hydrogen atom undergo _____ Condensation

15. Methyl alcohol reacts with CH_3COOH in the presence of H_2SO_4 to give _____.

III. Match the following

16. $k[CH_3Br][OH^-]$	a) Phenol
17. Cyclooctatetraene	b) S_N2
18. Carbolic acid	c) 8 pi electrons
19. Clemmensen reagent	d) 2 amino Benzoic acid
20. Anthranilic acid	e) Zn/Hg and HCl

SECTION - B (10X2=20)

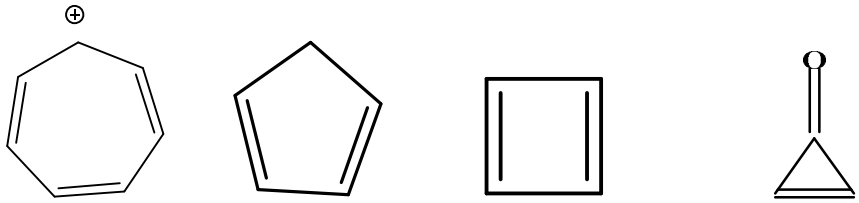
Answer any **TEN** Questions

21. What is solvolysis?

22. What are nucleophiles? Give examples?

23. What is resonance energy?

24. Which of the following species are aromatic?

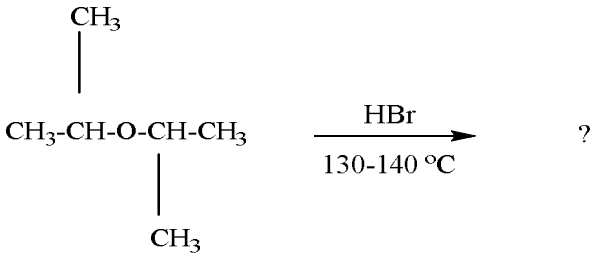


(a) (b) (c) (d)

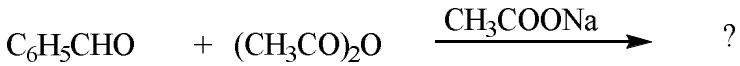
25. Predict the products



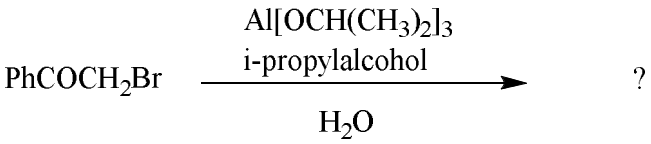
26. Predict the product

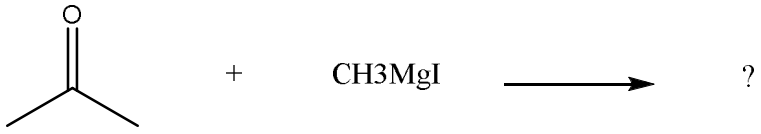


27. Arrange water, ethanol and phenol in increasing order of acidity and give reason.

28. 

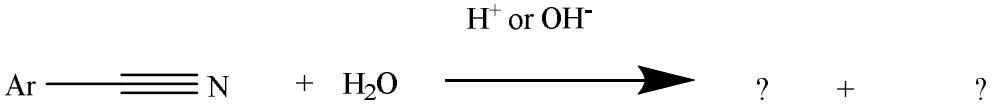
29.



30. 

31. Give the expression for acidity constant?

32. Predict the products



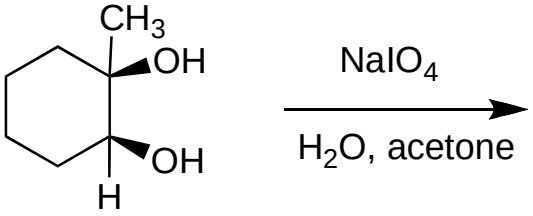
SECTION - C (5X7=35)

Answer any **FIVE** Questions

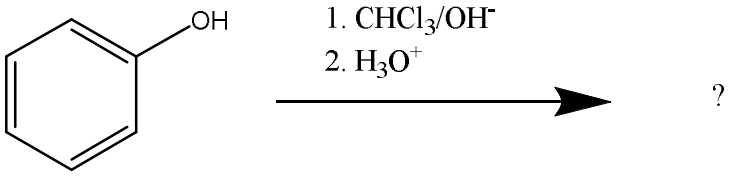
33. a) Prove that the SN2 reaction is stereospecific with mechanism? **(3)**
    b) Compare SN_1 and SN_2 reaction mechanisms with examples. **(2)**
    c) State the conditions of best leaving group. Give some Examples. Arrange the order of leaving group of halide ions and compare it with the order of basicity. **(2)**

34. a) What is an aromatic, anti-aromatic and non-aromatic compound? Explain with examples? **(3)**
    b) Write the mechanism of sulphonation? **(2)**
    c) Explain the formation of aniline from a haloarene with potassium amide using benzyne mechanism. **(2)**

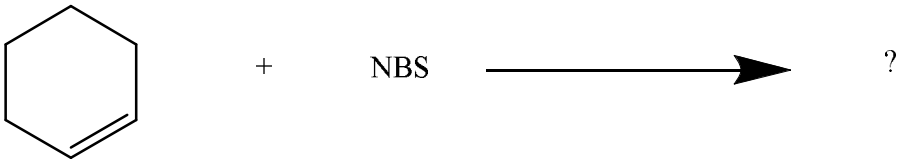
35. a) Predict the product. **(2)**



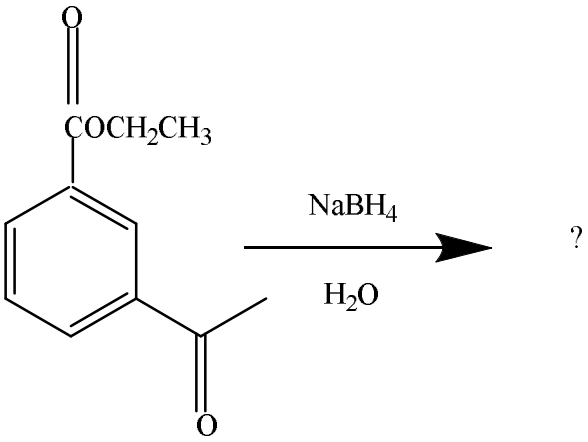
    b) Predict the product and show all the steps involved. **(3)**



    c) Predict the product. **(2)**

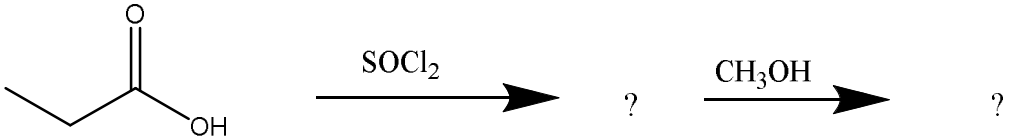


36. a) Give the mechanism of Benzoin condensation. **(3)**
    b) Predict the product. **(2)**



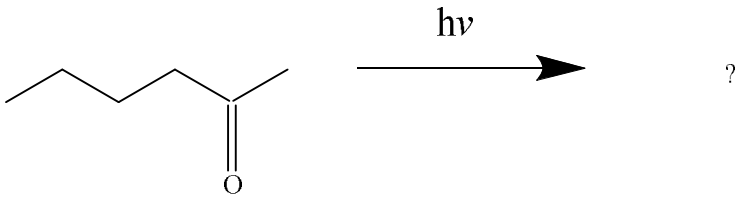
    c) Explain Norrish Type I reaction. **(2)**

37. a) How do you prepare succinic and adipic acid. **(2)**
    b) Find the products (2)



    c) Give a Suitable example of Hell – Volhard Zelinsky reaction? **(3)**

38. a) Write the mechanism and find the product. **(3)**

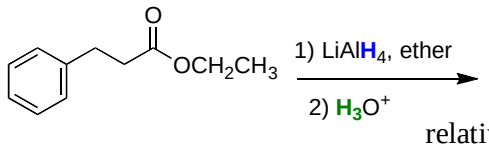


b) How will you know whether a given OH group is alcoholic or phenolic in nature? **(2)**

c) The bond length of >C=O in carboxylic acid is slightly larger than that in aldehydes and ketones. Why?

39. a) Why acetaldehyde gives aldol condensation. While formaldehyde does not. Explain.**(3)**

b) Predict the product. **(2)**



c) Arrange the relative reactivities of following alkyl halides under S_N2 (2)

CH_3-X, $(CH_3)_2CH$-X, CH_3CH_2-X and $(CH_3)_3C$-X

Esters

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